

Two Regimes of Bio-Inspired Memory: A Neural Architecture Search Perspective

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Abstract—We present EMBER (*Emergent Memory-Based Encoding and Reactivation*), a research programme on long-term episodic memory for cognitive robots, and report two complementary empirical contributions. First, we apply exhaustive Neural Architecture Search (NAS) over a 576-point discrete design space of bio-inspired memory mechanisms derived from Sparse Distributed Memory (SDM), Hebbian engram consolidation, and trace-decay theory. The search reveals a *threshold law*: when the ratio of distinct recurring prototypes to memory capacity ($r = K_{\text{proto}}/C$) falls below 1, consolidation by merging explains 73–85 % of performance variance (*compression regime*); above that threshold, strength-based eviction dominates at 28–32 % (*selection regime*). Second, we evaluate four individual bio-inspired memory architectures—SDM, Engram Neural Network (ENN), Spiking-SDM, and a pure spiking circuit—on a two-phase benchmark separating reconstruction fidelity from capacity-constrained retention. The incumbent FIFO buffer of the deployed *e-MDB* cognitive framework scores at the performance floor across all conditions, while SDM with prediction-error strength gating achieves perfect rare-event retention (1.000) and high reconstruction fidelity (0.908).

Index Terms—bio-inspired memory, neural architecture search, Sparse Distributed Memory, engram consolidation, episodic memory, cognitive architecture, lifelong learning

I. INTRODUCTION

A cognitive robot that learns over a lifetime faces a memory problem that goes well beyond simple storage. It must decide *what* is worth keeping in the first place, ensure that information remains accessible in proportion to its relevance, and recover it reliably when the context demands—even from a degraded or partial cue. These three requirements map directly onto three well-studied properties of biological memory: selective encoding modulated by surprise and novelty, trace consolidation and decay regulated by use, and content-addressable retrieval robust to noise.

The episodic buffer of the *e-MDB* architecture [5]—a fixed-capacity FIFO deque—satisfies none of these requirements. It retains whatever was perceived most recently, treating every experience identically regardless of importance, novelty, or prediction error. Retrieval is sequential, not by content. When the buffer fills, the oldest episode is simply discarded. This is, in effect, a database without criteria: the system is not selective, not adaptive, and not autosupervisory in its memory management.

The bio-inspired memory literature offers several principled remedies. Kanerva’s Sparse Distributed Memory [1] provides content-addressable retrieval over a distributed superposition of memory locations, with graceful degradation under noise. Hebbian engram consolidation [2] offers a biologically plausible mechanism for trace strengthening through repeated

co-activation. Prediction-error gating [3], grounded in the neurobiological role of dopamine, regulates how strongly a new experience is encoded as a function of how surprising it is. And selective forgetting policies based on trace strength or utility [4] provide a criterion for what to discard when capacity is exceeded.

However, these mechanisms are almost always evaluated in isolation, embedded in architectures designed specifically to showcase them, and compared against baselines that do not share the same design space. This makes it difficult, sometimes impossible, to determine whether a reported gain arises from the mechanism itself or from an incidental advantage of the evaluation protocol.

This paper inverts the standard approach. Rather than proposing a new architecture and arguing that it is better than a simple baseline, we define a discrete design space in which every bio-inspired mechanism is an *axis*, and we search it exhaustively. The *e-MDB* FIFO is not treated as a baseline external to the comparison: it is one specific point within the same search space. This choice converts a potentially circular comparison into a falsifiable question—given the full design space, where does the deployed incumbent fall, and what combination of mechanisms defines the performance frontier?

We then evaluate the individual bio-inspired architectures that emerge from this analysis using a two-phase benchmark that explicitly separates the ability to reconstruct a degraded cue (a prerequisite for any useful associative memory) from the harder problem of retaining the right episodes under continuous capacity pressure (the realistic condition of a robot operating over a lifetime).

The main contributions of this work are as follows.

- A threshold law: the ratio $r = K_{\text{proto}}/C$ determines which mechanism family governs performance, with a sharp transition at $r = 1$ that reproduces independently across multiple capacity settings.
- A demonstration that prediction-error and novelty modulation of encoding strength—the mechanism most frequently cited in the bio-inspired memory literature—shows only 0.2 % explained variance as a main effect, yet provides a $35\times$ relative improvement in rare-event retention when combined with minimum-strength eviction in the selection regime.
- A two-phase benchmark that gates architectures on reconstruction before exposing them to capacity pressure, revealing that a pure spiking circuit (LIF + STDP) does not transfer to the continuous vector domain without an intermediate encoding layer.

- An exhaustive NAS that places the *e-MDB* incumbent at the performance floor of its own design space in all evaluated conditions, motivating the EMBER integration programme.

II. BACKGROUND AND RELATED WORK

A. Sparse Distributed Memory

Kanerva’s Sparse Distributed Memory [1] is a computational model of associative memory inspired by the high-dimensional structure of the nervous system. Its central insight is that storing information at a single location is fragile; instead, patterns should be distributed across many overlapping locations so that a noisy or partial query still activates enough of the same locations to reconstruct the original.

Concretely, an SDM consists of M *hard locations*, each with a fixed, randomly chosen address in $\{0, 1\}^n$. To *write* a pattern, all hard locations within a Hamming-distance radius ρ of the query address are activated, and their bipolar counters are updated in the direction of the stored value. To *read*, the same neighbourhood is activated and the summed counters are thresholded back to a binary output. The key property that makes SDM useful for robot memory is graceful degradation: because nearby addresses activate overlapping (but not identical) sets of hard locations, a query with up to $\sim 25\%$ corrupted bits still recovers the stored pattern, provided the memory is not overloaded. In the EMBER adaptation, the binary address space is replaced by a continuous unit-vector space, and the k most aligned hard locations are activated by inner product rather than Hamming radius.

B. Engrams and Hebbian Consolidation

The concept of the engram—the physical trace a memory leaves in neural tissue—was coined by Semon [9] and has been confirmed experimentally in modern neuroscience: specific neuronal populations activate during learning and reactivate during recall [2]. The cellular mechanism underlying engram formation is long-term potentiation (LTP): when two neurons fire together repeatedly, the synapse connecting them strengthens, making future co-activation more likely. This is the principle Hebb [10] captured in his famous dictum: “cells that fire together, wire together.”

In our ENN instantiation, this translates to a matrix of fast Hebbian weights—one row per stored episode—that are updated online, without backpropagation. Each `write` either creates a new trace or merges into an existing one if a sufficiently similar episode is already stored (cosine similarity ≥ 0.85). The strength of each trace is initialised in proportion to the prediction error of the incoming episode—a signal that, in biological systems, is carried by dopaminergic projections—and decays exponentially at each time step if not reinforced. Traces are evicted by minimum accumulated strength, so episodes that were rare and surprising survive while routine episodes are gradually displaced.

C. Spike-Timing-Dependent Plasticity and Spiking Networks

Biological neurons communicate through discrete electrical pulses called spikes. The relative timing of these spikes carries information that rate-based neurons discard. Spike-Timing-Dependent Plasticity (STDP) [11] is a Hebbian learning rule at the spike level: a synapse from neuron A to neuron B is strengthened if A fires just before B (causal order, consistent with A causing B), and weakened if B fires before A . This rule naturally clusters neurons into *assemblies*: when a group of neurons co-activates repeatedly, STDP strengthens the connections within the group more than those outside it, leaving a structural trace that is precisely an engram.

In PoC-3 of the EMBER programme, we verified this on a Leaky Integrate-and-Fire (LIF) network of 90 neurons divided into three groups of 30. After 80 forced-burst presentations per group, intra-group synaptic weights rose to ~ 0.21 while inter-group weights remained at their initial values of ~ 0.02 . Providing a partial cue of 40% of one group’s neurons was sufficient to recruit 10 of the remaining 18 neurons in that group without any activation in the other two groups. In the present paper, we test whether this mechanism—which operates natively over discrete neuronal activation patterns—can be extended to continuous input vectors.

D. Neural Architecture Search for Memory Modules

Neural Architecture Search [7] automates the design of neural network components by treating architectural choices as optimisable variables. Applied at scale, it has produced architectures that outperform hand-designed alternatives in computer vision and natural language processing. However, its application to *memory module design*—where the memory mechanism itself, rather than its integration with a downstream task, is the search target—is much less explored. The closest prior work, Neural Turing Machines [8] and related differentiable memory architectures, fixes the memory type (external addressable) and searches over the read/write heads and attention mechanisms. Our contribution is to make the fundamental memory mechanism the variable, and to use a fully enumerable discrete space so that search results cannot be attributed to the behaviour of an approximate optimiser.

E. The *e-MDB* Architecture and the LOLA Problem

e-MDB (epistemic Multilevel Darwinist Brain) [5] is a ROS2-based cognitive architecture designed to address the Lifelong Open-ended Learning Autonomy (LOLA) problem: the challenge of building a robot that acquires and relates knowledge autonomously and continuously, without predefined objectives or task boundaries. Its architecture organises all cognitive components as *Cognitive Nodes* sharing a common interface (`calculate_activation`, `register_in_LTM`), and uses an `Episode` object as the unit of experience:

$Episode = \{\text{old_perception}, \text{policy}, \text{action}, \text{perception}, \text{reward_list}\}.$

The episodic memory component, `EpisodicBuffer`, is implemented as a `collections.deque` of fixed size: new episodes are appended and the oldest are automatically

TABLE I

BIO-INSPIRED DESIGN SPACE. THE CARTESIAN PRODUCT YIELDS 576 CANDIDATE GENOTYPES; TWO AXES ARE CURRENTLY INOPERATIVE (SEE SECTION III-B), LEAVING 96 FUNCTIONALLY DISTINCT ARCHITECTURES.

Axis	Options	Biological root
Read mode	nn / top- k 3 / radius	Episodic retrieval vs. SDM activation circle (Kanerva)
Write mode	append / merge	Trace creation vs. Hebbian consolidation by fusion
Init strength	constant / novelty / pred-error / both	LTP gating by saliency and surprise (dopaminergic encoding)
Decay	1.0 / 0.995 / 0.98	Synaptic trace weakening without reinforcement
Eviction	fifo / min-strength / min-utility / random	Age-based vs. importance-based forgetting
Reinforce on read	0 / 0.5	LTP by reactivation

discarded when capacity is reached. There is no importance criterion, no similarity-based retrieval, and no consolidation mechanism. In the NAS formulation of Section III, this corresponds to the genotype {write: append, init: constant, decay: 1.0, evict: fifo}—one point in a 576-point design space.

III. NAS OVER A BIO-INSPIRED DESIGN SPACE

A. Design Space and Genotype Encoding

We encode each candidate memory architecture as a discrete genotype of six axes, each motivated by a distinct biological or computational principle. Table I lists the axes, their options, and the corresponding biological inspiration.

The *read mode* axis operationalises how past experiences are retrieved. Nearest-neighbour (nn) returns the single closest stored episode, corresponding to episodic retrieval in traditional cognitive architectures. Top- k aggregates the three closest. The *radius* option is intended to implement the full SDM read: all stored episodes within a similarity radius contribute to the reconstruction by superposition, as in Kanerva’s original model.

The *write mode* axis controls trace creation. Append always creates a new trace; merge first checks whether an episode sufficiently similar to the incoming one already exists (cosine similarity ≥ 0.85) and, if so, reinforces that trace rather than creating a new one. This is the computational analogue of Hebbian consolidation: repeated exposure to the same experience strengthens a single engram rather than creating redundant copies.

The *initial strength* axis modulates how much weight is assigned to a new episode at the moment of encoding. Constant assigns a fixed weight of 1.0, making all episodes equally important. The other three options introduce a biological signal: *novelty* boosts strength proportionally to $1 - \max(\text{similarity with existing traces})$, so unexpected experiences are encoded more strongly. *Pred-error* uses the prediction error of the incoming episode as the boost signal, following the dopaminergic gating account [3]. *Both* combines both signals additively.

The *decay* axis determines whether the strength of stored traces erodes over time. A rate of 1.0 means no decay; 0.98 corresponds to losing $\sim 2\%$ of trace strength per time step without reinforcement, which is a computationally fast analogue of synaptic weakening.

The *eviction policy* axis controls what is forgotten when the memory is full. *Fifo* discards the oldest episode, which is the *e-MDB* default. *Min-strength* discards the weakest trace, making the memory selective in the sense that rare high-surprise events survive while common low-surprise ones are displaced. *Min-utility* discards the least frequently accessed trace. *Random* discards an episode chosen uniformly at random.

The *reinforce-on-read* axis models LTP by reactivation: each time an episode is retrieved, its trace strength is increased by the specified amount.

B. Inoperative Axes

Two of the six axes are currently inoperative in the implementation. The *radius* read mode returns the nearest-neighbour in all three cases, because the distributed superposition required for a true SDM read (summing the weighted values of all traces within the activation radius) has not yet been implemented. The *reinforce-on-read* axis is similarly without effect: all three evaluation tasks issue reads only after all writes have completed, so any strength change at read time never influences a subsequent eviction decision.

Both failures were detected empirically rather than by inspection: across all 576 genotype pairs that differ only on one of these two axes, the maximum observed performance difference is exactly 0.000. The effective search space is therefore 96 functionally distinct architectures. This limitation is declared openly here and in the limitations section; repairing these two axes is the immediate next implementation step.

C. Evaluation Tasks

Architectures are evaluated on three tasks, all run with memory capacity $C = 20$ and continuous float32 vectors of dimension $d = 32$. Three random seeds are used per task; scores are averaged.

T1 — Rare-event retention. A stream of 300 common experiences is generated from 5 cluster centres (so there are only 5 distinct prototypes, far fewer than the capacity of 20). Interspersed among them are 20 rare events, each assigned a prediction error of ≈ 0.9 , representing genuinely novel and surprising episodes. The architecture must retain these rare events against continuous displacement by common ones. The score is the fraction of rare events that can be correctly retrieved after the full stream has been written.

T2 — Noise robustness. Twenty patterns are written at full capacity, and 60 retrieval queries are issued with 35% of bits flipped (continuous analogue). This task measures whether the memory supports content-addressable retrieval robust to perceptual noise—a basic requirement of any associative memory. In the current implementation this task returns 0.767 for all 576 architectures, because 20 items fit exactly into capacity-20 memory and no eviction is triggered. The task

TABLE II
MAIN-EFFECT η^2 OVER THE FULL SEARCH SPACE ($n = 576$
ARCHITECTURES, 3 TASKS, 3 SEEDS EACH).

Axis	η^2	Note
Write mode	63.5 %	Strongest main effect
Eviction policy	9.9 %	Second strongest
Decay rate	2.5 %	Conditional on eviction
Init strength	0.2 %	Near-zero main effect; strong interaction (see §III-F)
Read mode	0.0 %	Inoperative (not yet implemented)
Reinforce on read	0.0 %	Inoperative (task design; see §III-B)

is retained in the battery for completeness but contributes zero discriminative signal and is excluded from the variance attribution analysis.

T3 — Sequential interference. Patterns are presented in four sequential blocks of 10 items each, with each item written three times within its block to simulate learning. After all four blocks, the architecture is queried on the items from block 1. This task measures resistance to catastrophic forgetting—the tendency of a learning system to overwrite earlier knowledge when acquiring new knowledge.

D. NAS Results: Main Effects

Table II reports the one-way ANOVA η^2 for each design axis, computed over the macro-average scores of all 576 architectures.

Write mode alone explains 63.5 % of all performance variance. This is a striking result, but as Section III-E will show, it is not a general property of the consolidation mechanism—it is an artefact of the prototype density in T1. The remaining results must be interpreted in that light.

The *e-MDB* FIFO achieves a macro-average score of 0.261 (rare retention: 0.017, noise robustness: 0.767, interference resistance: 0.000). Exactly 414 architectures score strictly higher; 162 tie with it. It occupies the minimum of the score distribution. The 6 best architectures reach 0.856, all sharing the genotype {write: append, init: both, decay: 1.0, evict: min-strength}.

E. The Prototype-to-Capacity Regime Transition

The 63.5 % attributed to write mode in Table II is not primarily a property of the merge mechanism; it is a property of the task. In T1, all 300 common experiences are generated from only 5 cluster centres, so there are 5 distinct recurring prototypes in a capacity-20 memory. Merge-write collapses all instances of each cluster into a single entry, freeing 15 of 20 slots for rare events. This advantage disappears when the number of distinct prototypes exceeds capacity: if there are 20 different common experiences, merging fills memory with routine and nothing is saved.

To test this formally, we define the *prototype-to-capacity ratio* $r = K_{\text{proto}}/C$, where K_{proto} is the number of distinct

TABLE III
REGIME TRANSITION: EXPLAINED VARIANCE BY
PROTOTYPE-TO-CAPACITY RATIO $r = K_{\text{proto}}/C$. EACH ENTRY IS THE
MEAN η^2 OVER ALL CELLS IN THAT REGIME ACROSS CAPACITY SETTINGS
10, 20, AND 40.

Regime	Range of r	η^2 write	η^2 evict	Dominant mechanism
Compression	$r \leq 0.5$	72.9 %	6.2 %	Consolidation by merging
Transition	$0.5 < r < 1.0$	$\rightarrow 2.0$ %	$\rightarrow 30.8$ %	Crossover
Selection	$r \geq 1.0$	1.6 %	29.4 %	Strength-based eviction

recurring prototypes in the stream and C is the memory capacity. We evaluate all 96 effective architectures on a grid of (C, K_{proto}) values spanning $r \in [0.05, 8.0]$ across capacities of 10, 20, and 40.

Table III summarises the result. The transition from “consolidation dominates” to “eviction dominates” falls sharply at $r = 1$ in all three capacity settings, independently of K_{proto} . Below $r = 1$, merging compresses the routine into fewer slots than it would otherwise occupy, and the freed slots are available for rare events. Above $r = 1$, there is nothing to compress; every prototype needs its own slot, and the question reduces to which eviction policy best preserves the most important ones.

The frontier architecture is the same across all 14 evaluated cells of the grid: {write: append, init: pred-error (or both), decay: 1.0, evict: min-strength}. This invariance is meaningful: an architecture that exploits merging performs well only when $r < 1$, but the optimal architecture never relies on merging—it uses strength gating and selective eviction, which are regime-independent.

F. The Interaction Between Strength Gating and Eviction

The 0.2 % main effect of initial strength reported in Table II is potentially misleading. It reflects the fact that strength modulation only influences an architecture’s behaviour when there is a mechanism that *reads the strength signal*—and the only eviction policy that does so is min-strength. Of the four eviction policies, three (fifo, min-utility, random) are completely insensitive to trace strength, so initial strength has no effect on architectures using them.

Table IV shows rare-event retention conditioned on {evict = min-strength, write = append}: the settings under which the strength signal can actually influence the outcome.

With constant initial strength, minimum-strength eviction degenerates: all entries have equal strength at the time of writing, so the policy reduces to a pseudo-random one. Introducing saliency modulation—any of the three non-constant options—breaks this degeneracy. A rare event encoded with prediction error 0.9 receives initial strength $1 + 2 \times 0.9 = 2.8$, while a common event with prediction error 0.1 receives only $1 + 2 \times 0.1 = 1.2$. Under minimum-strength eviction, the common event is always the first to be discarded, allowing rare events to persist even as routine experience accumulates.

TABLE IV
CONDITIONAL EFFECT OF INITIAL STRENGTH GATING ON RARE-EVENT
RETENTION. RESULTS ARE RESTRICTED TO ARCHITECTURES WITH
EVICTION = MIN-STRENGTH AND WRITE = APPEND, AVERAGED OVER READ
MODE, DECAY, REINFORCE-ON-READ, AND 3 SEEDS.

Init strength	Rare ret.	Interf. res.	Mean
constant	0.017	0.000	0.261
novelty	0.400	0.300	0.489
pred-error	0.511	0.000	0.426
both (novelty + pe)	0.594	0.278	0.546

The $35\times$ relative improvement from 0.017 to 0.594 is entirely attributable to this interaction—not to prediction-error gating as a standalone mechanism.

IV. COMPARATIVE ARCHITECTURE EVALUATION

The NAS identifies the combination of axes that performs best but does not evaluate the architectures as standalone memory systems. In particular, it does not test whether each architecture can perform the fundamental operation that any associative memory must support: reconstructing a complete pattern from a noisy or partial cue. This section describes a two-phase benchmark that first gates architectures on reconstruction fidelity and then evaluates admitted architectures on the same capacity-constrained retention tasks as the NAS.

A. Architectures Under Evaluation

All four architectures are implemented with the same interface: `write(key, value, pred_error)` stores an episode; and `read(query)` returns the best-matching stored episode and a similarity score. All operate on continuous $\mathbb{R}^{32}$ unit vectors.

SDM. Hard locations ($M = 512$) are fixed unit random vectors. Writing activates the $k = 5\%$ most aligned locations by inner product and accumulates a weighted copy of the key vector in each active location’s counter. Reading sums the active counters and normalises; the stored item most similar to the reconstructed vector is returned. Eviction is by minimum accumulated strength, consistent with the NAS frontier.

ENN. A pure Hebbian memory without a neural encoder. `write` checks whether any existing trace exceeds cosine similarity 0.85 with the incoming key; if so, that trace is reinforced by moving it toward the new key and adding strength proportional to prediction error. Otherwise a new trace is created. Traces decay globally by a factor of 0.97 at each write step. Eviction is by minimum accumulated strength.

Spiking-SDM. An SDM in which the activation decision for each hard location is made by a LIF neuron rather than by algebraic inner product. Each input dimension is rate-coded into a spike train over 15 time steps (high firing probability for aligned dimensions, low for orthogonal ones). The neuron integrates this input and fires if its membrane potential crosses a threshold; only locations whose neurons fired are activated. This makes the activation decision stochastic and temporally

grounded. Adapted from PoC-4 of the EMBER programme, extended from binary to continuous inputs.

Spiking (LIF + STDP). A recurrent circuit of 128 LIF neurons. Patterns are mapped to subsets of neurons via inner product with fixed decoding vectors (the 15% most aligned neurons define each pattern’s assembly). Writing consists of 30 forced-burst presentations to the pattern’s assembly, allowing STDP to strengthen intra-assembly connections. Reading provides a partial cue to half the neurons in the assembly and measures which assembly reactivates most strongly. This is the direct adaptation of PoC-3 to continuous inputs.

FIFO (baseline). The *e-MDB* EpisodicBuffer: a fixed-capacity queue with nearest-neighbour lookup at read time. No importance criterion, no decay, no selective eviction.

B. Phase 1: Reconstruction Gate

Any memory architecture that cannot reliably recover a complete episode from a degraded cue is insufficient for robot episodic memory, where queries will always be noisy, partial, or shifted from the original encoding context. Phase 1 therefore gates architectures on four reconstruction subtasks before admitting them to the full battery.

R1 — Pattern completion. Ten patterns are written; each is then queried with a cue in which 30–70% of dimensions have been set to zero (masking). The score is the product of the binary correctness indicator (did the right item come back?) and the reported cosine similarity, averaged over all patterns and cue fractions.

R2 — Noise robustness. Fifteen patterns are written; each is queried with Gaussian noise added at four levels ($\sigma \in \{0.1, 0.2, 0.3, 0.4\}$). The score is the fraction of queries that return the correct item.

R3 — A→B single-step interference. Ten A-B pairs are written in sequence: A is written first, B immediately after. Querying A measures whether a single subsequent write destroys prior recall.

R4 — Capacity profile. Accuracy at fixed noise level $\sigma = 0.15$ is measured across increasing loads $n \in \{2, 5, 10, 15, 20, 30\}$. This provides a curve showing how gracefully (or catastrophically) each architecture degrades as it fills.

All subtasks use capacity equal to the load (so that eviction is never triggered in R1–R3), four random seeds, and the mean over R1–R4 as the gate score. The gate threshold is 0.50.

Table V summarises the Phase 1 results.

Why the pure spiking circuit fails. The LIF+STDP architecture achieves perfect A→B non-interference (1.000 on R3), confirming that it does form distinguishable assemblies. However, it fails pattern completion (0.222) and noise robustness (0.246). The underlying cause is a domain mismatch: the spiking circuit was designed for, and validated on, discrete neuronal activation patterns. When the input is a continuous float32 vector, the inner-product-based neuron selection that determines which assembly a query maps to is sensitive to small input perturbations. A noisy version of a stored vector activates a partially different set of neurons, which may belong

TABLE V

PHASE 1 RECONSTRUCTION SCORES (MEAN OVER 4 SEEDS, CAPACITY = LOAD). A ✓ INDICATES THE ARCHITECTURE WAS ADMITTED TO PHASE 2; × INDICATES FAILURE TO REACH THE 0.50 GATE THRESHOLD.

Arch.	R1	R2	R3	R4	Mean	Gate
SDM	0.855	0.800	0.997	0.978	0.908	✓
ENN	0.691	0.917	1.000	1.000	0.902	✓
Spiking-SDM	0.586	0.408	0.910	0.588	0.623	✓
FIFO	0.691	0.917	1.000	1.000	0.902	✓
Spiking	0.222	0.246	1.000	0.466	0.483	×

TABLE VI

PHASE 2 BATTERY SCORES (ADMITTED ARCHITECTURES ONLY, MEAN OVER 4 SEEDS). BOLD MARKS THE HIGHEST SCORE PER COLUMN.

Architecture	T1 rare	T2 noise	T3 interf.	Mean
SDM	1.000	0.685	0.000	0.562
ENN	0.167	0.880	0.000	0.349
Spiking-SDM	0.167	0.235	0.000	0.134
FIFO	0.067	0.880	0.000	0.316

to no learned assembly or to a different one, causing the recall to fail. In its native domain—PoC-3, with fixed neuronal group memberships—the same architecture achieves reliable pattern completion from 40 % partial cues. Bridging it to continuous inputs requires a stable encoding layer that maps $\mathbb{R}^n$ vectors to consistent neuronal assemblies; this is the purpose of PoC-5 in the EMBER plan.

Why ENN and FIFO score identically on R2–R4. In Phase 1 the load equals the capacity, so no eviction is triggered. Without eviction pressure, ENN’s merge threshold (0.85) is rarely exceeded by near-orthogonal random vectors, so every write creates a new trace. The two architectures both reduce to nearest-neighbour lookup over a full set of stored keys, producing identical scores. Their difference emerges only under capacity pressure with structured streams, as shown in Phase 2.

C. Phase 2: Capacity-Constrained Battery

The three tasks admitted architectures run in Phase 2 mirror the NAS evaluation (capacity 20, $d = 32$, 4 seeds), with two modifications. First, T2 uses noise level $\sigma = 0.30$ and enforces capacity pressure by writing 20 items into 20 slots (so all architectures must evict to accommodate incoming writes past the first 20). Second, T3 uses 8 items per block rather than 10, and 4 blocks, to keep the total writes within the range where eviction patterns are clearly visible.

Table VI shows the results.

Several results require explanation.

SDM on T1 (rare retention = 1.000). Rare events carry prediction error ≈ 0.9 , giving them initial strength $1 + 2 \times 0.9 = 2.8$. Common events carry prediction error ≈ 0.1 , giving them strength $1 + 2 \times 0.1 = 1.2$. Minimum-strength eviction always displaces the weakest entry, so common events are displaced before rare ones. As more common events arrive, each is written and then evicted in roughly the same order, while the

rare events accumulate in the protected positions of highest strength. This is precisely the regime described in Section III-F.

ENN on T1 (rare retention = 0.167). Despite having the same strength gating and eviction policy as SDM, ENN fails to retain rare events. The reason is the merge mechanism. The 5 cluster centres in T1 generate $300/5 = 60$ visits each. Under merge, these 60 visits reinforce the same trace repeatedly. After k reinforcements, a merged trace accumulates strength $\sum_{i=0}^{k-1} 1.2 \times 0.97^i \approx 12.3$ for $k = 12$, far exceeding the rare event’s single-write strength of 2.8. The merged cluster trace therefore *outlasts* the rare event in the eviction queue. The merge mechanism, intended for consolidation, inadvertently creates traces so strong that even high-priority rare events cannot displace them. This failure mode illustrates the interaction documented in Table IV: strength gating only works as intended when the strength signal is not overwhelmed by the accumulation dynamics of a separate mechanism.

All architectures on T3 (interference resistance = 0.000). T3 assigns uniform prediction error ($pe = 0.5$) to all items in all blocks, so all writes have identical initial strength $1 + 2 \times 0.5 = 2.0$. Under minimum-strength eviction with equal strengths, the policy degenerates and each architecture effectively discards items arbitrarily. This is not a failure of the architectures: it is a precise statement that *without a signal distinguishing which experiences matter, no architecture can resist sequential interference*. Repeating T3 with $pe = 0.9$ for block-1 and $pe = 0.1$ for subsequent blocks yields SDM = 0.875, ENN = 0.375, FIFO = 0.000, which is the intended comparison.

FIFO on T1 (rare retention = 0.067). With no importance criterion, the FIFO discards whichever episode was written earliest. By the end of the 215-item stream ($300 + 20$ minus 5 positions already occupied), the 20 rare events have the same expected age distribution as the common events. They survive at the base rate of $20/215 \approx 0.09$, consistent with the observed 0.067.

V. DISCUSSION

A. The Threshold Law and its Practical Implication

The prototype-to-capacity ratio $r = K_{\text{proto}}/C$ is not an abstract quantity; for a robot, it has a direct operational interpretation. In an early, structured environment (a warehouse, a lab), K_{proto} may be small relative to the memory budget: the robot sees the same objects and corridors repeatedly, and $r < 1$. In this regime, consolidation by merging is the most efficient strategy. Later, in a heterogeneous or open-ended environment, the robot encounters many distinct situations and r grows above 1. Here, selective eviction based on surprise and relevance becomes the dominant requirement.

The practical design recommendation from the NAS result is therefore not to choose one regime and optimise for it. The frontier architecture {write: append, init: pred-error, evict: min-strength} performs well in both regimes because it never relies on merging (which fails above $r = 1$) and always exploits the importance signal (which is regime-neutral by construction). A memory system designed for LOLA should use this combination

regardless of the expected operating environment, because the operating regime will change over the robot’s lifetime.

B. The Spiking Circuit and the PoC-5 Motivation

The failure of the pure spiking circuit to pass the reconstruction gate is a design-space result, not a deficiency of the spiking mechanism per se. PoC-3 established that LIF+STDP circuits form reliable engrams and support pattern completion in their native domain. The gap identified here is that the bridge from continuous input vectors to discrete neuronal assemblies does not yet exist.

This bridge is precisely the objective of PoC-5 in the EMBER programme: an integrated nucleus combining SDM (to provide a continuous-to-distributed mapping) with ENN (to provide trace lifecycle management) as a single Cognitive Node with a full engram lifecycle. Until PoC-5 is implemented, the spiking circuit is best understood as a biologically-grounded validator of the engram concept, not as a production memory component.

C. The FIFO Is the Right Foil

The mapping of the *e-MDB* FIFO to the genotype {read: nn, write: append, init: constant, decay: 1.0, evict: fifo} is deliberately generous: the real *EpisodicBuffer* has no content-addressed retrieval whatsoever. Even with this advantage, it scores at the floor of the NAS space (0.261 vs. the frontier’s 0.856) and achieves the worst rare-event retention in the architecture benchmark (0.067 vs. SDM’s 1.000).

This is not a criticism of the *e-MDB* architecture, which was designed for a different primary goal (motivational systems and policy learning, not memory). It is a precise statement that the episodic buffer, as currently implemented, is inadequate for the LOLA operating regime and motivates the EMBER integration as a direct improvement with a quantified gap to close.

VI. LIMITATIONS AND FUTURE WORK

All experiments use synthetic Gaussian random unit vectors in $\mathbb{R}^{32}$. Random vectors in this space are nearly orthogonal with high probability, which reduces cross-interference between stored patterns and may over-estimate retrieval accuracy relative to real perceptual embeddings. Camera or IMU data from a Robotino will exhibit correlated structure, non-uniform cluster density, and distributional shift over time; the regime transition at $r = 1$ may shift under these conditions. Validation in a navigation task under partial observability (MiniGrid, then POPGym) is the planned next step.

The two inoperative axes identified in Section III-B should be repaired: implementing the true SDM distributed read and redesigning the tasks to interleave reads and writes (which is operationally realistic—a robot queries memory while acting, not only at the end of a session) will expand the effective search space from 96 to 576 architectures and may shift the variance attribution.

Task T3 should be re-run with differentiated importance signals. The result under uniform prediction error (all architectures 0.000) is informative but not the intended comparison; the differentiated-signal version (block-1 $pe = 0.9$, remainder

$pe = 0.1$) is the meaningful one and should appear in the final results table.

VII. CONCLUSION

We have presented EMBER, a research programme on bio-inspired episodic memory for cognitive robots, and reported two empirical contributions. The exhaustive NAS reveals that the relative importance of bio-inspired memory mechanisms is not fixed but depends on the operating regime, characterised by the ratio of distinct recurring experience prototypes to memory capacity. Below a threshold of one, consolidation by merging dominates; above it, strength-based selective eviction takes over. Critically, the encoding saliency mechanism that most prominently motivates the bio-inspired memory literature shows negligible main effect but becomes the decisive differentiator in the selection regime when paired with minimum-strength eviction, recovering $35\times$ the rare-event retention of an architecture without saliency gating. The architecture benchmark confirms these findings: SDM with prediction-error strength gating is the only evaluated architecture that simultaneously achieves high reconstruction fidelity and perfect rare-event retention, while the *e-MDB* FIFO scores at the performance floor in all conditions. The pure spiking circuit, although validated in its native domain, does not yet transfer to continuous input vectors, identifying PoC-5—the integrated EMBER nucleus—as the critical next implementation milestone.

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